Fusion Categories and SymTFT: Homework #3

Fusion Rings and Fusion Categories
February 10, 2026

Problem 1: Deligne product

(a) Say we have two fusion rings, with label sets L_1, L_2 and fusion coefficients $N_{c_1}^{a_1 b_1}$ for $a_2, b_2, c_2 \in L_2$ and $N_{c_2}^{a_2 b_2}$ for $a_2, b_2, c_2 \in L_2$, respectively. **Check** that $L = L_1 \times L_2$ – which we will write as $L = \{a \boxtimes b\}_{a \in L_1, b \in L_2}$ – gives the free Z-module on L the structure of a fusion ring with respect to the multiplication on L given by $(a_1 \boxtimes a_2) \times (b_1 \boxtimes b_2) = \sum_{c_1 \in L_1, c_2 \in L_2} N_{c_1}^{a_1 b_1} N_{c_2}^{a_2 b_2} c_1 \boxtimes c_2$. (4 points)
We will see that this categorifies to to a product of categories called the Deligne product, and that this operation can be interpreted as a stacking of physical theories.

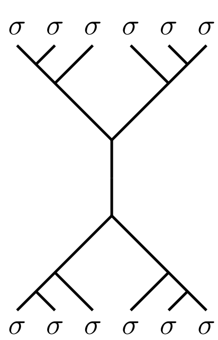
(b) **Fill in the blanks** in the fusion table for the fusion ring Fib $\boxtimes$ Ising.

	$1 \boxtimes 1$	$1 \boxtimes \sigma$	$1 \boxtimes \psi$	$\tau \boxtimes 1$	$\tau \boxtimes \sigma$	$\tau \boxtimes \psi$
$1 \boxtimes 1$	$1 \boxtimes 1$	$1 \boxtimes \sigma$	$1 \boxtimes \psi$	$\tau \boxtimes 1$	$\tau \boxtimes \sigma$	$\tau \boxtimes \psi$
$1 \boxtimes \sigma$	$1 \boxtimes \sigma$					
$1 \boxtimes \psi$	$1 \boxtimes \psi$					
$\tau \boxtimes 1$	$\tau \boxtimes 1$					
$\tau \boxtimes \sigma$	$\tau \boxtimes \sigma$					
$\tau \boxtimes \psi$	$\tau \boxtimes \psi$					

(c) **Explain** why $d_{a \boxtimes b} = d_a d_b$ in any Deligne product of fusion rings, where d_a is the Frobenius-Perron dimension.[1] (2 points)

Problem 2: Admissible labelings

Enumerate the admissible labelings of this diagram using the Ising fusion rules. Assume all edges are oriented going down the page.



[1] Recall from lecture that this notation d_a for $a \in L$ is typically reserved for the quantum dimension of an object (rather than the Frobenius-Perron dimension in either a unitary or spherical fusion category but that we're co-opting it anyway.

Problem 3: Morphisms and Trivalent graphs

Consider a fusion category $\mathcal{C}$ with simple objects $\{a, b, \dots\}$, whose fusion rules are encoded by structure constants N_{ab}^{c}. Assume fusion multiplicities are finite.

Consider the vector space

$$\mathrm{Hom}(a \otimes b,\ a \otimes b).$$

1. **Basis states.** Describe a basis for $\mathrm{Hom}(a \otimes b,\ a \otimes b)$ in terms of admissibly labeled trivalent graphs. Your answer may be given as diagrams or as a description of all allowed intermediate labels.
2. **Dimension.** Show that the dimension of this vector space is

$$\dim \mathrm{Hom}(a \otimes b,\ a \otimes b) \ = \ \sum_{c} N_{ab}^{c} N_{c}^{ab}.$$

3. **Special case.** Assume that all fusion multiplicities satisfy $N_{ab}^{c} \in \{0, 1\}$. Explain why $\dim \mathrm{Hom}(a \otimes b,\ a \otimes b)$ is equal to the number of distinct fusion channels of $a \otimes b$.